

## **JAVA WITH DSA-D12**

### **HOTEL MANAGEMENT SYSTEM**

```
import java.util.*;
class Room
{
    private int roomNumber;
    private String roomType;
    private double pricePerDay;
    private boolean booked;
    private String customerName;
    private int days;
    public Room(int roomNumber, String roomType, double pricePerDay)
    {
        this.roomNumber = roomNumber;
        this.roomType = roomType;
        this.pricePerDay = pricePerDay;
        this.booked = false;
        this.customerName = "";
        this.days = 0;
    }
    public int getRoomNumber()
    {
        return roomNumber;
    }
    public String getRoomType()
    {
        return roomType;
    }
    public double getPricePerDay()
    {
        return pricePerDay;
    }
    public boolean isBooked()
    {
        return booked;
    }
    public String getCustomerName()
    {
        return customerName;
    }
    public int getDays()
    {
        return days;
    }
}
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    }
    public void bookRoom(String customerName, int days)
    {
        booked = true;
        this.customerName = customerName;
        this.days = days;
    }
    public void checkOut()
    {
        booked = false;
        customerName = "";
        days = 0;
    }
    public void displayRoom()
    {
        System.out.println();
        System.out.println("Room Number : " + roomNumber);
        System.out.println("Room Type : " + roomType);
        System.out.println("Price Per Day : " + pricePerDay);
        System.out.println("Status : " + (booked ? "Booked" : "Available"));
        if(booked)
        {
            System.out.println("Customer : " + customerName);
            System.out.println("Days Booked : " + days);
        }
    }
}
class Hotel
{
    private ArrayList<Room> rooms;
    public Hotel()
    {
        rooms = new ArrayList<>();
    }
    private Room findRoom(int roomNumber)
    {
        for(Room room : rooms)
        {
            if(room.getRoomNumber() == roomNumber)
            {
                return room;
            }
        }
    }
    return null;
}

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    }
    public boolean roomExists(int roomNumber)
    {
        return findRoom(roomNumber) != null;
    }
    public void addRoom(int roomNumber, String roomType, double pricePerDay)
    {
        if(!(roomType.equalsIgnoreCase("Single") ||
roomType.equalsIgnoreCase("Double") || roomType.equalsIgnoreCase("Deluxe")))
        {
            System.out.println("Invalid Room Type!");
            return;
        }
        if(pricePerDay <= 0)
        {
            System.out.println("Invalid Price!");
            return;
        }
        rooms.add(new Room(roomNumber, roomType, pricePerDay));
        System.out.println("Room Added Successfully.");
    }
    public void viewAllRooms()
    {
        if(rooms.isEmpty())
        {
            System.out.println("No Rooms Available.");
            return;
        }
        for(Room room : rooms)
        {
            room.displayRoom();
        }
    }
    public void viewAvailableRooms()
    {
        boolean found = false;
        for(Room room : rooms)
        {
            if(!room.isBooked())
            {
                room.displayRoom();
                found = true;
            }
        }
    }

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        if(!found)
        {
            System.out.println("No Available Rooms.");
        }
    }
    public void searchRoom(int roomNumber)
    {
        Room room = findRoom(roomNumber);
        if(room == null)
        {
            System.out.println("Room Not Found.");
        }
        else
        {
            room.displayRoom();
        }
    }
    public void bookRoom(int roomNumber, String customerName, int days)
    {
        Room room = findRoom(roomNumber);
        if(room == null)
        {
            System.out.println("Room Not Found.");
            return;
        }
        if(room.isBooked())
        {
            System.out.println("Room Already Booked.");
            return;
        }
        if(days <= 0)
        {
            System.out.println("Invalid Number of Days.");
            return;
        }
        room.bookRoom(customerName, days);
        System.out.println("Room Booked Successfully.");
    }
    public void checkOut(int roomNumber, String customerName)
    {
        Room room = findRoom(roomNumber);
        if(room == null)
        {
            System.out.println("Room Not Found.");
        }
    }

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        return;
    }
    if(!room.isBooked())
    {
        System.out.println("Room is Already Available.");
        return;
    }
    if(!room.getCustomerName().equals(customerName))
    {
        System.out.println("You did not book this room.");
        return;
    }
    double bill = room.getPricePerDay() * room.getDays();
    System.out.println("\nHOTEL BILL ");
    System.out.println("Customer Name : " + customerName);
    System.out.println("Room Number : " + room.getRoomNumber());
    System.out.println("Room Type : " + room.getRoomType());
    System.out.println("Days Stayed : " + room.getDays());
    System.out.println("Price Per Day : " + room.getPricePerDay());
    System.out.println("");
    System.out.println("Total Bill : " + bill);
    System.out.println("");
    room.checkOut();
    System.out.println("Check Out Successful.");
}
public void deleteRoom(int roomNumber)
{
    for(int i = 0; i < rooms.size(); i++)
    {
        Room room = rooms.get(i);
        if(room.getRoomNumber() == roomNumber)
        {
            if(room.isBooked())
            {
                System.out.println("Cannot Delete a Booked Room.");
                return;
            }
            rooms.remove(i);
            System.out.println("Room Deleted Successfully.");
            return;
        }
    }
}
System.out.println("Room Not Found.");
}

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```

public void viewMyBookings(String customerName)
{
    boolean found = false;
    for(Room room : rooms)
    {
        if(room.getCustomerName().equals(customerName))
        {
            room.displayRoom();
            System.out.println("Days Stayed : " + room.getDays());
            found = true;
        }
    }
    if(!found)
    {
        System.out.println("No Bookings Found.");
    }
}
public void totalRooms()
{
    System.out.println("Total Rooms : " + rooms.size());
}
public void totalAvailableRooms()
{
    int count = 0;
    for(Room room : rooms)
    {
        if(!room.isBooked())
        {
            count++;
        }
    }
    System.out.println("Available Rooms : " + count);
}
}
public class Main
{
    static Scanner sc = new Scanner(System.in);
    static Hotel hotel = new Hotel();
    public static void main(String[] args)
    {
        while(true)
        {
            System.out.println("\nHOTEL MANAGEMENT SYSTEM");
            System.out.println("1. Admin Login");

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        System.out.println("2. User Login");
        System.out.println("3. Exit");
        System.out.print("Enter Your Choice : ");
        int choice = sc.nextInt();
        switch(choice)
        {
            case 1:
                adminLogin();
                break;
            case 2:
                userLogin();
                break;
            case 3:
                System.out.println("Thank You! We Hope You Enjoyed the Stay!");
                System.exit(0);
            default:
                System.out.println("Invalid Choice.");
        }
    }
}

static void adminLogin()
{
    sc.nextLine();
    System.out.print("Enter Admin Username : ");
    String username = sc.nextLine();
    System.out.print("Enter Admin Password : ");
    String password = sc.nextLine();
    if(username.equals("admin") && password.equals("1234"))
    {
        System.out.println("Admin Login Successful.");
        adminMenu();
    }
    else
    {
        System.out.println("Invalid Username or Password.");
    }
}

static void userLogin()
{
    sc.nextLine();
    System.out.print("Enter Your Name : ");
    String customer = sc.nextLine().trim();
    if(customer.isEmpty())
    {

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        System.out.println("Name Cannot Be Empty.");
        return;
    }
    userMenu(customer);
}
static void adminMenu()
{
    while(true)
    {
        System.out.println("\nADMIN MENU");
        System.out.println("1. Add Room");
        System.out.println("2. View All Rooms");
        System.out.println("3. Search Room");
        System.out.println("4. Delete Room");
        System.out.println("5. Total Rooms");
        System.out.println("6. Logout");
        System.out.print("Enter Your Choice : ");
        int choice = sc.nextInt();
        switch(choice)
        {
            case 1:
                System.out.print("Enter Room Number : ");
                int roomNo = sc.nextInt();
                if(hotel.roomExists(roomNo))
                {
                    System.out.println("Room Number Already Exists!");
                    break;
                }
                sc.nextLine();
                System.out.print("Enter Room Type (Single/Double/Deluxe) : ");
                String type = sc.nextLine();
                System.out.print("Enter Price Per Day : ");
                double price = sc.nextDouble();
                hotel.addRoom(roomNo, type, price);
                break;
            case 2:
                hotel.viewAllRooms();
                break;
            case 3:
                System.out.print("Enter Room Number : ");
                hotel.searchRoom(sc.nextInt());
                break;
            case 4:
                System.out.print("Enter Room Number : ");

```



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        hotel.deleteRoom(sc.nextInt());
        break;
    case 5:
        hotel.totalRooms();
        break;
    case 6:
        System.out.println("Admin Logged Out.");
        return;
    default:
        System.out.println("Invalid Choice.");
    }
}
}
static void userMenu(String customer)
{
    while(true)
    {
        System.out.println("\nUSER MENU");
        System.out.println("1. View Available Rooms");
        System.out.println("2. Book Room");
        System.out.println("3. Check Out");
        System.out.println("4. Search Room");
        System.out.println("5. View My Bookings");
        System.out.println("6. Logout");
        System.out.print("Enter Your Choice : ");
        int choice = sc.nextInt();
        switch(choice)
        {
            case 1:
                hotel.viewAvailableRooms();
                break;
            case 2:
                System.out.print("Enter Room Number : ");
                int roomNo = sc.nextInt();
                System.out.print("Enter Number of Days : ");
                int days = sc.nextInt();
                hotel.bookRoom(roomNo, customer, days);
                break;
            case 3:
                System.out.print("Enter Room Number : ");
                hotel.checkOut(sc.nextInt(), customer);
                break;
            case 4:
                System.out.print("Enter Room Number : ");

```

```
        hotel.searchRoom(sc.nextInt());
        break;
    case 5:
        hotel.viewMyBookings(customer);
        break;
    case 6:
        System.out.println("Logged Out Successfully.");
        return;
    default:
        System.out.println("Invalid Choice.");
    }
}
}
```